

Product Requirements Document: BudgetMind

AI Cashflow Forecaster & Crisis Advisor

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1. Project Overview & Business Value

BudgetMind is a predictive financial health dashboard designed specifically for Small and Medium Enterprises (SMEs). Unlike traditional accounting software that focuses on historical record-keeping, BudgetMind is a forward-looking engine. Its primary mission is to eliminate the "Cash Flow Gap"—the leading cause of 82% of small business failures—by providing visibility into liquidity crises up to 90 days in advance. By integrating Google Gemini 1.5 Pro, BudgetMind acts as a "CFO in a Box," transforming complex financial data into actionable, human-readable mitigation strategies.

2. User Personas

Alex: The High-Growth Freelancer

- **Role:** Independent Creative Director.
- **Challenges:** Inconsistent payment cycles from multiple clients; difficulty tracking when large tax payments or equipment upgrades will impact bank balances.
- **Goals:** To know exactly when it is "safe" to spend on business growth and to avoid overdraft fees during dry spells.

Sarah: The Small Agency Owner

- **Role:** Founder of a 10-person digital marketing agency.
- **Challenges:** Managing payroll, office rent, and vendor payments across fluctuating project-based revenue; lack of time to perform complex "what-if" modeling for new hires.
- **Goals:** To maintain a 3-month cash runway and understand the immediate financial impact of hiring a new developer or losing a major retainer.

3. User Stories

ID	User Role	Requirement
US.1	Freelancer ▾	As a freelancer, I want to upload my monthly bank export (CSV)
US.2	Agency Owner ▾	As an owner, I want the AI to highlight why my cash will dip below zero in July
US.3	Agency Owner ▾	As an owner, I want to ask "What if I hire a senior designer at \$6k/month?"
US.4	All Users ▾	As a user, I want to toggle a "Crisis Mode"
US.5	All Users ▾	As a user, I want to receive early warning alerts via email/dashboard

4. Functional Features

Cashflow Forecasting

- **Mechanism:** Aggregates time-series data from CSV/financial logs.
- **Output:** Interactive 90-day chart showing "Running Balance" vs. a user-defined "Safe Operating Threshold".

AI Crisis Advisor (The "Secret Sauce")

- **Engine:** Powered by Google Gemini 1.5 Pro.
- **Logic:** Analyzes historical and projected patterns to identify specific shortfall causes (e.g., tax dates, seasonal dips) and suggests natural language solutions.

Decision Simulator

- **Interface:** Natural Language Processing (NLP) chat interface.
- **Functionality:** Allows users to input hypothetical scenarios (hiring, equipment purchase, losing a client) to see real-time adjustments to the 90-day forecast.

5. Technical Architecture

- **Frontend:** Streamlit for rapid, interactive Python-based UI development.
- **AI Engine:** Google Generative AI SDK (Gemini 1.5 Pro) for advanced reasoning and prompt engineering.
- **Data Layer:** Pandas and NumPy for high-performance time-series aggregation and financial logic.
- **Visualization:** Altair/Plotly for dynamic, responsive charting of financial trajectories.

6. Feature Roadmap

Phase 1: MVP - The Foundation

- Manual CSV Ingestion (Bank/Xero/QuickBooks exports).
- Core Forecasting Engine (90-day linear projections).
- Basic Altair visualization of running balances.

Phase 2: AI Advisor Integration

- Integration of `gemini_client.py` for automated pattern analysis.
- "Crisis Mode" toggle for stress-testing resilience.
- Automatic identification of "Safe Operating Threshold" breaches.

Phase 3: Advanced Simulation & NLP

- Implementation of the text-driven "What-if" Decision Simulator.
- Prompt engineering for industry-specific financial advice.
- Exportable "Financial Health Reports" for bank/loan applications.

Phase 4: Ecosystem & Automation

- Direct API Integrations (Xero, QuickBooks, FreshBooks).
- Real-time bank feed connection via Plaid/Salt Edge.
- Automated notification system for 21-day "Early Warnings."

7. Key Success Metrics

- **Manual Effort Reduction:** Decrease spreadsheet reconciliation time by 5 hours/week.
- **Lead Time:** Provide a minimum 21-day "Early Warning" before liquidity drops below the threshold.
- **Financial Impact:** Aim for a 15-20% acceleration in cash inflow through AI-suggested payment optimizations.